File Sharing Design Document

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1.0 Product Breif

As the App is yet to be named i shall refer to it as File Transfer Application (FTA). FTA is a file Sharing Program for local networks. The functionality will work simularly to Apples Airdrop except its multiplatform.

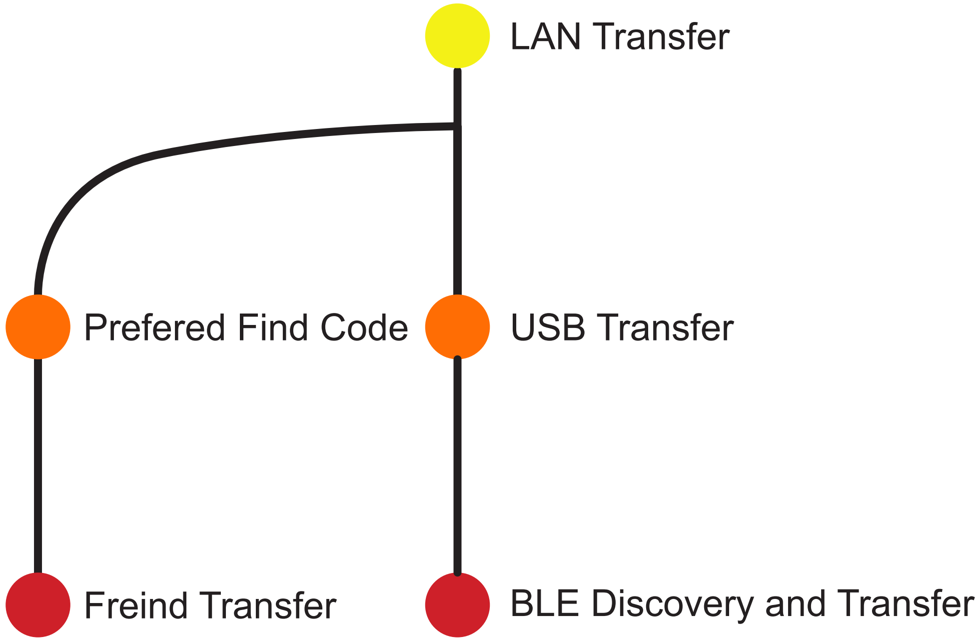
Some proposed features for this app include. LAN File share, BLE Detection with wifi direct file share, USB tranfer for mobile to pc -> ultrafast. We will also implment custom icons, find codes, transfer level system and a contacts location mode. This will improve functionality for multiple devices.

Identified limitations.
The biggest limitation to this software is IOS. IOS is infamous for there poor developer treatment and completly locked down system. This means BLE Detection will not work without the app being open on ios, meanig ios will be the most limited platform for this tool.

Important notes

Philosiphy - keep complexity without sacrificing usability. The application should have no learning curve to the average person while allowing people to do alot.

1.1 Feature Map



2.0 Protocol

The Protocol will be the same across all platform this allows for similar to identical functionality across all platfofrms running the app. This is essentialy the communication layer between all platforms so information transfer.

The Protocol will first start with a magic number this is to differentiate itself from other devices over a network allowing us to easily identify a packet created by the FTA App. The next will be a 4-character code this is the communication code. Every communication on the network for this app will use this as the start of a message

USE A HMAC for Broadcast

Magic Number	ID Code	Data
FTA	EXMP	...

The following sections will now cover the different Codes and there meaning and a break down of when messages will be sent. The data section is organised by first in list

CODE: FIND

Data: Find Code, N, H

Use: this is used to create the find code which is a unique id on a network. This is sent out with the generated or selected find code and if we recive no response we know the find code is avaliable.

CODE: RFIN

Data: rec Find Code

Use: this is the response to the FIND command. This includes the find code recived from the origional FIND message. This tells the other user that this find code exists on the network so you need to generate another one.

CODE: BEAT

Data: Find Code, S, Username, icon

Use: this is the response to the FIND command. This includes the find code recived from the origional FIND message. This tells the other user that this find code exists on the network so you need to generate another one.

CODE: DELE

Data: Find Code

Use: This will use ip matching based on the first beat recived from this find code,

EXTRA:

FIND n & h and Beat S

so what we will do is generate a random number, plus hash the S (secret) + the Number, to get H now this should give us h if they match we know that this is the person that origionaly sent the find

2.1 Protocol Direct

Magic Number	ID Code	Data
FTB	EXMP	...

CODE: REQS

Data: Find Code, Username, filename, base, modulus, public key

Use: this is the request Command, used to send a file transfer request it also contains diffe-helmen data. Including the requesting computers final secret.

CODE: ACCP

Data: public key, Find Code, Username

Use: this is the Accept Command it is used to accept an incoming request

CODE: CANC

Data: public key, Find Code, Username

This can be sent at anytime to close a conection and delete the transfered content. This is just to inform the sender or reciver that the transfer was canceled

Below Messages are Encrypted

CODE: HEAD

Data: Find Code, Username, file count

Use: this is the intial command sent by the sender, it is to tell the reciver the ammount of files

CODE: FILE

Data: Find Code, Username, file id, filename, filesize, blockcount, checksum (SHA-512)

Use: this is the head of the current file. So when this is set we create a blank file, the checksum is saved until the final block is recived then we compare the final output to the checksum if we fail (which we shouldnt) we can request the file again using ERRR

CODE: DATA

Data: Find Code, File id, Block num, data count, data, checksum (SHA-256)

Use: this is the request Command, used to send a file transfer request it also contains diffe-helmen data. Including the requesting computers final secret.

CODE: FAIL

Data: Find Code, Username, File id, Block num

Use: This is used in 2 senarios, 1st is if the checksum at the end of data, or if we belive that the file has been transfered and we are still missing blocks. (note this does not indicate failure to transfer file but just a failure to find a block so send it again)

CODE: PASS

Data: Find Code, Username, Fileid

Use: this will send until the aknl command is recived but it just tells the sender to start the next file or to close the connection if we are in the pass loop for 30 seconds exit and keep the recived file

CODE: DONE

Data: Find Code, Username, Fileid

Use: This is used by us to check if the file has finished sending if we are uncertain if we get no response form the sender we can assume it has been

CODE: ERRR

Data: Find Code, Username, Fileid

Use: this is a drastic command, this is the total error command, only send if the final checksum failed even if all the data checksums passed, this will initiate a file resend where we send the same file again, because of corruption

CODE: AKNL

Data: Find Code, Username, file id

Use: this is the response to pass, pass will be sent every second until we get this response, only if the there is more than 1 file we are transfering, after this we send the next file header

CODE: EXIT

Data: Find Code, Username

Use: the last command this just tells the other client to disconect the other client should auto disconect after 30 seconds without communication, in case of unexpected internet outage or unexpected sender error. Or if we equal the file count in head

Precausions:

when the file header is sent we create the exact file size on reciving, however the file could be auto opened, the issue is that we dont want this, to midigate this all files will end with .tmp or .temp until the transfer is finished then that is removed

3.0 Product Designs

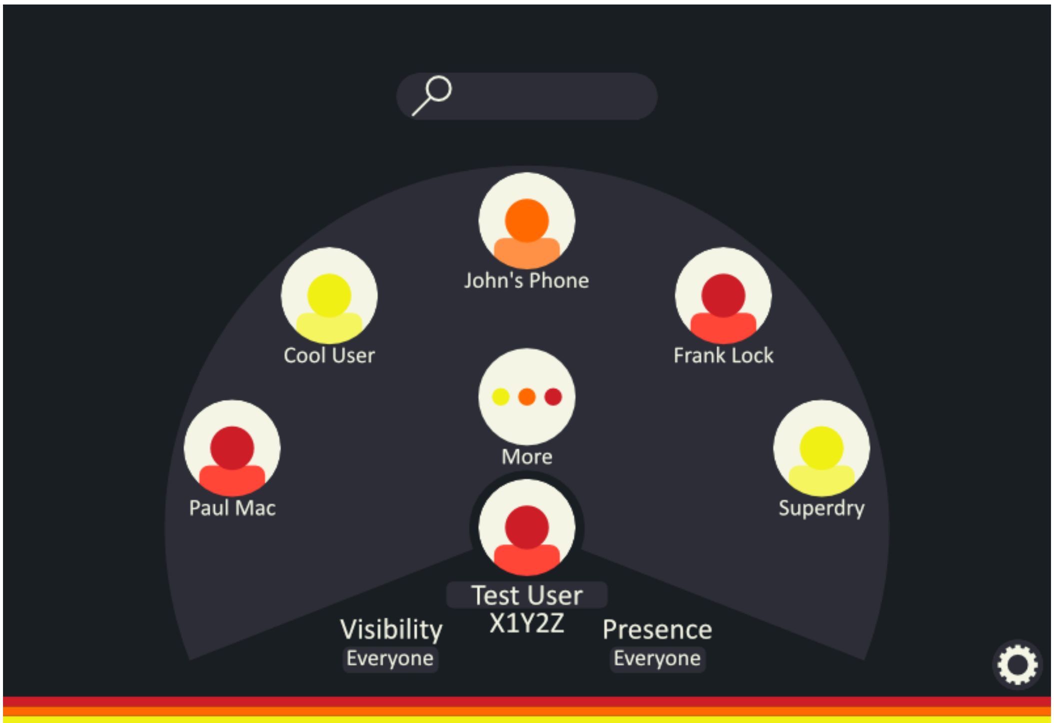
Colour Palette:

The Colour Palette is described in the Book of Colour V1

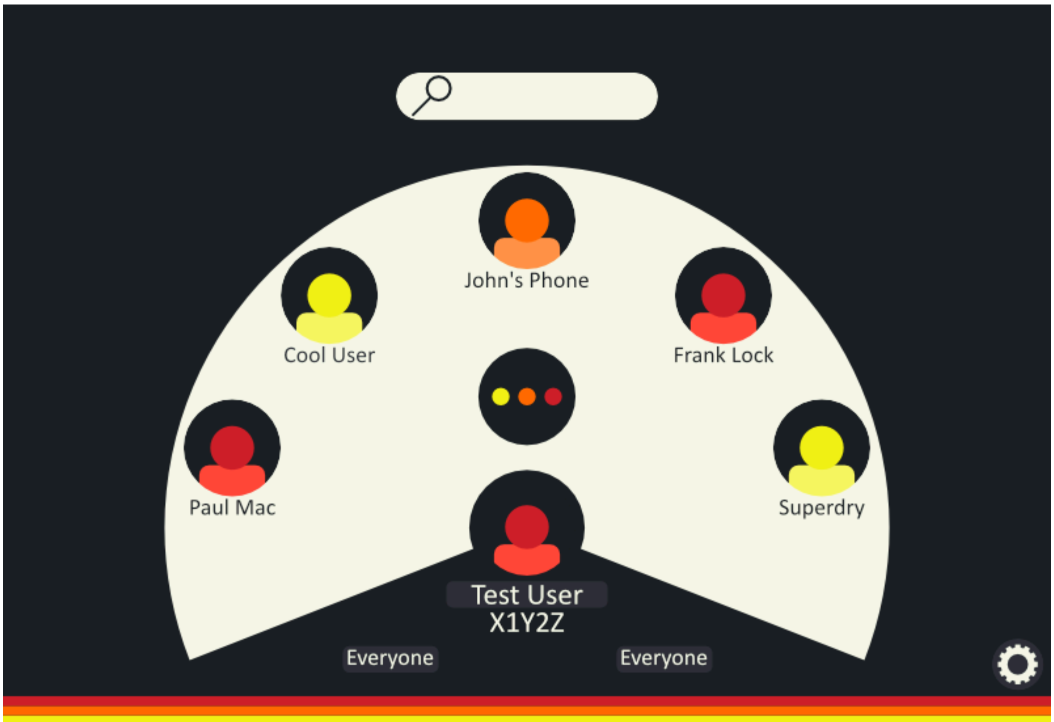
Design’s are Broken in to light and dark mode

First Designs

Rev 1




John's Phone
Cool User
Frank Lock
More
Paul Mac
Superdry
Test User
Visibility
Everyone
Presence
Everyone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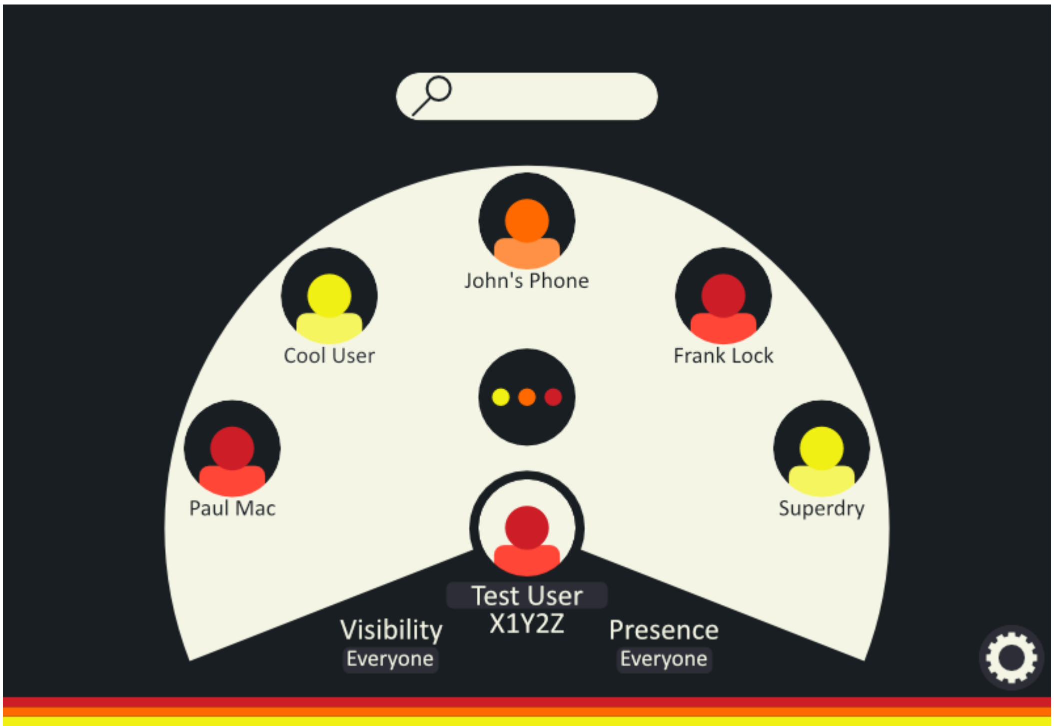
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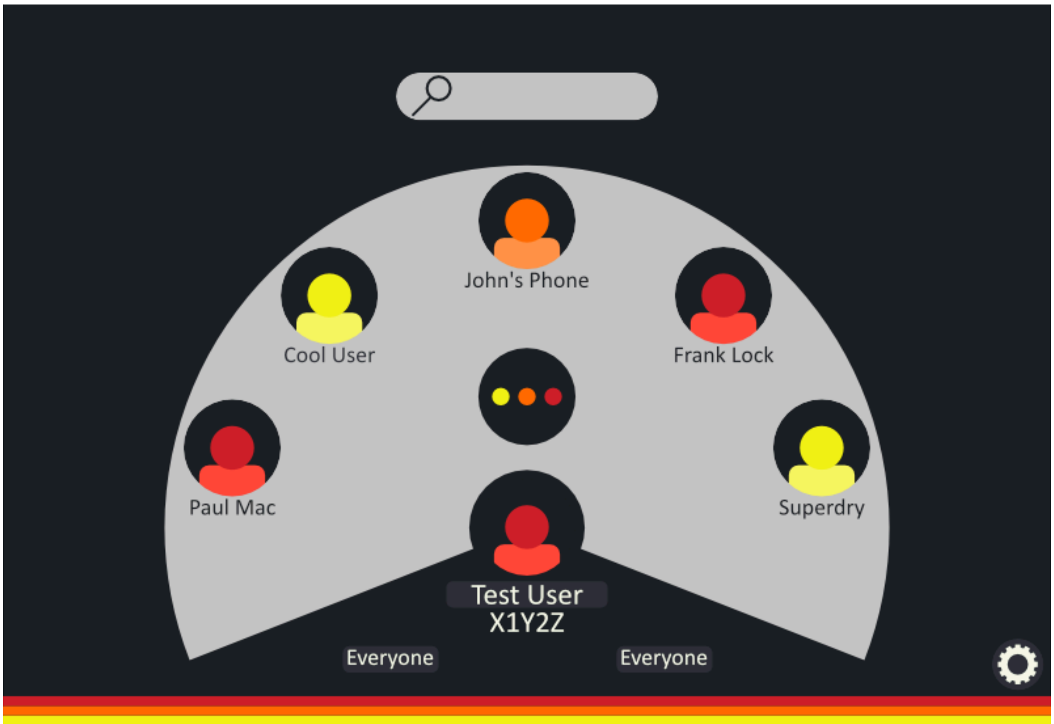
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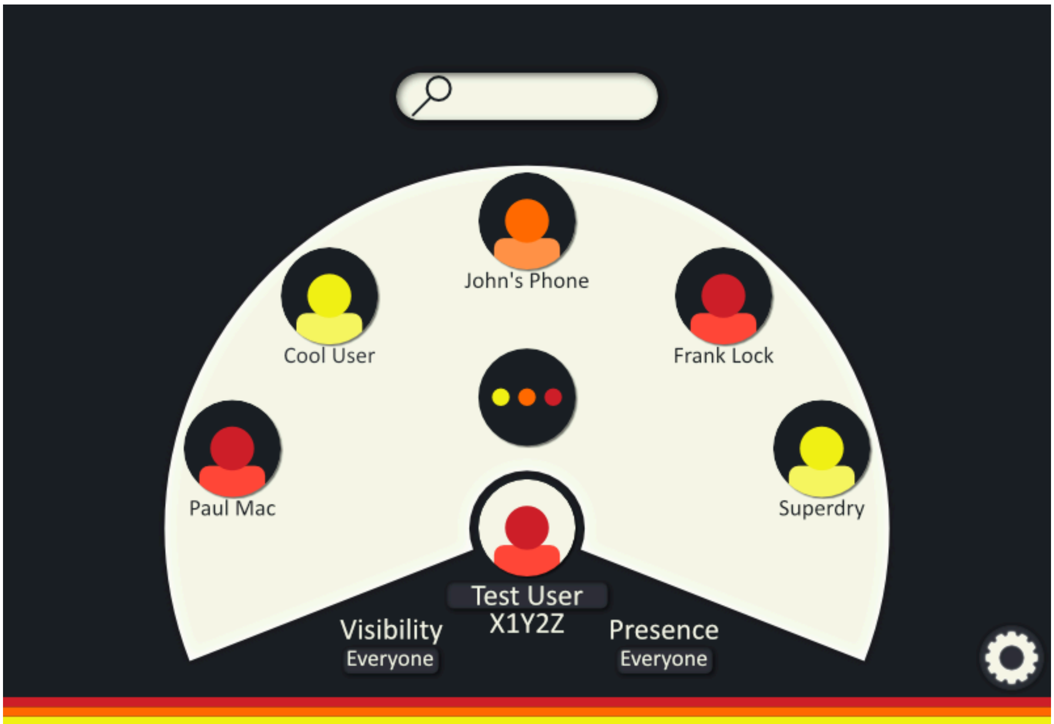
Rev 2


John's Phone
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X1Y2Z
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Rev 3


Paul Mac
Paul Mac
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Paul Mac
Paul Mac
Paul Mac
Default Na...
X1Y2Z